

examination, keyword searching, timeline, and web artifact extraction. It can produce detailed data about deleted files, online activity and user behaviour, which helps investigators

recreate events and find hidden evidence. It has a modular architecture and includes some strong reporting features for recording the results of an investigation as well.

Tools for analysis of Android apps

Let's take a quick look at the key open source tools that can be used to perform the analysis and digital forensics of Android apps.

Static analysis tools

Tool	URL	Application
Apktool	https://apktool.org	Decompiles and recompiles APKs (Android Package Kits) allowing for resource extraction and modification.
JD-GUI	java-decompiler.github.io	A graphical tool for decompiling .class files from Java applications.
JADX	github.com/skylot/jadx	Decompiles APK files to Java source code.

Dynamic analysis tools

Tool	URL	Application
Frida	frida.re	A dynamic instrumentation toolkit for testing and modifying APK behaviour.
Objection	github.com/sensepost/objection	Used for runtime security assessment of mobile applications without needing root.

Online APK scanners

Tool	URL	Application
VirusTotal	www.virustotal.com/	Uploads APKs to scan for malware using multiple antivirus engines.
MobSF (Mobile Security Framework)	github.com/MobSF/Mobile-Security-Framework-MobSF	A comprehensive mobile app security analysis tool that supports both static and dynamic analysis.

Reverse engineering platforms

Tool	URL	Application
Ghidra	ghidra-sre.org	A software reverse engineering framework from the NSA in the US, supporting APK analysis.
Androguard	github.com/androguard/androguard	A Python-based tool for analysing Android applications.



Figure 4: APKTool for reverse engineering of Android apps

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